



NATIONAL INSTITUTE OF TECHNOLOGY AGARTALA

DIGITAL ELECTRONICS PROJECT

Alpha-numeric 14 Segment Display using Logic Gates

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Submitted to :Dr. Yogesh Kumar Sariya

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Objective

The objective of this project is to **design a basic alphabet input system using only OR gates and a 14-segment alphanumeric display**, without the use of microcontrollers or programmable devices. The system allows users to press switches corresponding to different alphabets (A–Z), and displays the selected character by illuminating the appropriate segments on the display using combinational logic built purely with OR gates.

Components

1 14 Segment Display

2 IC 7432

3 IC 7404

4 Toggle switch

Working Principle

Introduction to Logic-Based Display CoWorking Principle –
Logic Control Without Microcontroller

Content:

- The project uses pure combinational logic to display alphabets on a 14-segment alphanumeric display.
- No microcontroller or programmable device is used — only switches and OR gates.
- Each alphabet is represented by a unique combination of segment lines (a–n).
- The logic determines which segments light up for each character based on the pressed key (switch).

 For example, pressing 'A' lights up segments a, b, c, e, f, g1, and g2.

Working Principle

Role of Switches in Character Input

Working Principle – Character Input via Switches

Content:

- Each switch corresponds to one alphabet character (e.g., A–Z).
- When a switch is pressed, it outputs a logic HIGH (5V or 1) signal.
- This HIGH signal is connected to the segments required for that specific character.
- The switches act as manual input signals in a keyboard-like setup.



Switch A → Segment lines a, b, c, e, f, g1, g2.

Working Principle

Segment Selection via OR Gate Logic

Working Principle – OR Gate-Based Segment Activation Content:

- Each of the 14 segments is connected to the output of an OR gate.
- Inputs to each OR gate come from all switches that require that segment.
- If any one of the switches is pressed, the OR gate passes a HIGH signal to the segment, turning it ON.
- This logic allows multiple characters to share segments without interference.

 Segment 'c' might be shared by A, B, and D — so all those switches connect to one OR gate for segment 'c'.

Working Principle

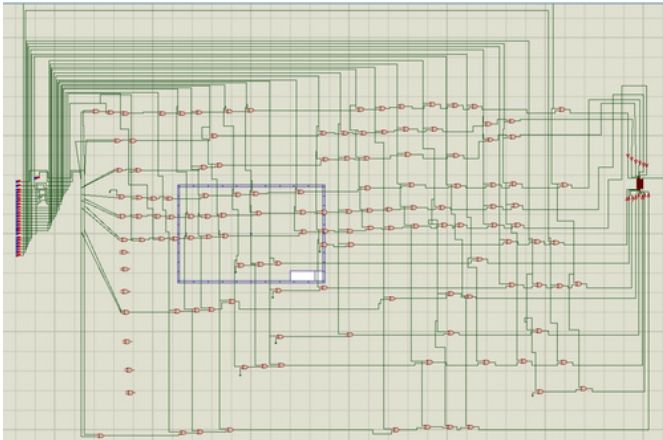
Output Display and Hardware Logic Summary

Working Principle – Final Display Output

Content:

- When a user presses any switch, the OR gates activate only the segments needed to form that character.
 - The 14-segment display lights up in the correct pattern for the pressed alphabet.
 - The system provides a clear, visible output based on simple hardware logic.
 - This makes it ideal for educational use, digital logic training, or low-cost embedded alternatives.
- ⚡ A real-time display system made using only gates and switches — no code, just logic!

Circuit Diagram



Procedure

- Design Segment Mapping:
- Identify which segments (a–n) need to be ON for each character (A–Z).
- Refer to the standard 14-segment display map for each letter.
- Assign Switches:
- Connect each switch to represent a unique alphabet character.
- Each switch will serve as the input source (logic HIGH) for a specific character.
- Create OR Gate Logic:
- For each segment, determine which characters need it.
- Connect the outputs of all relevant switches to the input of an OR gate for that segment.
- Repeat this for all 14 segments (a–n).
- Connect OR Gates to Display:
- Route the outputs of all OR gates to their corresponding segment pins on the 14-segment display.
- Use current-limiting resistors between the gates and display pins.
- Power Up the Circuit:
- Use a 5V DC regulated power supply.
- Ensure proper grounding and clean connections.
- Test Each Switch:
- Press individual switches and verify that the corresponding character appears correctly on the display.

Future Scope

Future Scope

Multi-Character Input System

Expand the project to allow entering multiple characters in sequence to form words or messages using a buffer/display array.

- Programmable Logic Integration

Replace basic OR gate logic with PLDs (Programmable Logic Devices) or FPGAs to allow dynamic character assignment.

Microcontroller-Based Upgrade

Integrate a microcontroller (like Arduino) to:

Control the display more efficiently

Add features like scrolling text, serial input, or backspace

Matrix-Based Keyboard Input

Build a proper keyboard matrix (like QWERTY) to reduce wiring complexity and improve user interface.

Voice-Controlled or Wireless Input

Interface Bluetooth modules or voice recognition modules for modern input methods.

Learning and Teaching Aid

- Use this hardware logic system as an educational tool for teaching digital electronics and logic design.

Conclusion

Conclusion

This project successfully demonstrates the use of basic digital logic to create a functional alphabet keyboard system.

By using only OR gates, switches, and a 14-segment display, it proves that character display is possible without any microcontroller or complex ICs.

The project reinforces the concepts of combinational logic, segment mapping, and hardware-based design.

It serves as an excellent learning tool for digital electronics, showcasing how logic gates can perform meaningful, visible tasks.

The simplicity and expandability of this system make it a great base for further innovations in display technology and logic-based design.